DIPLOMA IN SOFTWARE ENGINEERING

Module:

DIGITAL LOGIC AND COMPUTER ORGANIZATION

Title of the Course work:

Smart Room Light Controller - Project Report

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Abstract

This report presents the design and development of a Smart Room Light Controller using digital logic gate ICs. The system aims to automatically control room lighting based on motion detection, ambient light level, and energy-saving preferences. It integrates a PIR sensor, LDR, toggle switch, and pushbutton to provide automated, manual, and energy-efficient control. The circuit was developed and simulated using Tinkercad, ensuring both functionality and cost-effectiveness. The final prototype successfully meets the specified objectives.

Acknowledgment

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Contents

Abstract 2

Acknowledgment 2

Introduction 4

Methodology 4

Discussion 5

Future Implementations 5

References 5

Gantt Chart 6

Introduction

In the modern world, automation plays a crucial role in energy conservation and convenience. This project focuses on building a smart room light controller using fundamental logic gates. The light turns ON only when it is dark and motion is detected, with additional modes for manual override and energy-saving. The goal is to develop a low-cost, efficient system using discrete logic gate ICs for educational and practical implementation.

Methodology

- **Tools Used**: Tinkercad simulator, 7404 (NOT), 7408 (AND), 7432 (OR), 7486 (XOR) ICs, PIR sensor, LDR, SPST switch, pushbuttons, resistors, LED.
- **Design Process**:
    1. Used LDR and a voltage divider to detect light levels.
    2. PIR sensor integrated to detect motion.
    3. Logic conditions implemented using:
        - 7404 for NOT operation (darkness detection)
        - 7408 for AND (dark + motion)
        - 7486 XOR for energy mode control
        - 7432 OR for combining auto and manual controls
    4. Manual override with pushbutton
    5. Toggle switch for energy-saving mode
- **Simulation**:
    - Designed and tested in Tinkercad
    - Verified logic outputs for different input combinations

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Discussion

The circuit effectively performs its intended function. The logical design ensures the light only activates under desired conditions. Each component was strategically placed to optimize performance and minimize unnecessary activation. Manual override and energy mode introduce flexibility. Challenges included managing IC pin configurations and ensuring signal synchronization in Tinkercad. These were resolved through careful pin mapping and simulation testing.

Future Implementations

- Upgrade to microcontroller-based system (e.g., Arduino) for more complex decision-making
- Add light intensity threshold calibration via potentiometer
- Incorporate real-time clock for time-based control
- Add mobile app for remote control via IoT

References

- Sedra, A. S., & Smith, K. C. (2014). *Microelectronic Circuits*. 7th ed. Oxford University Press.
- All About Circuits. (2023). *Logic Gates Overview*. https://www.allaboutcircuits.com

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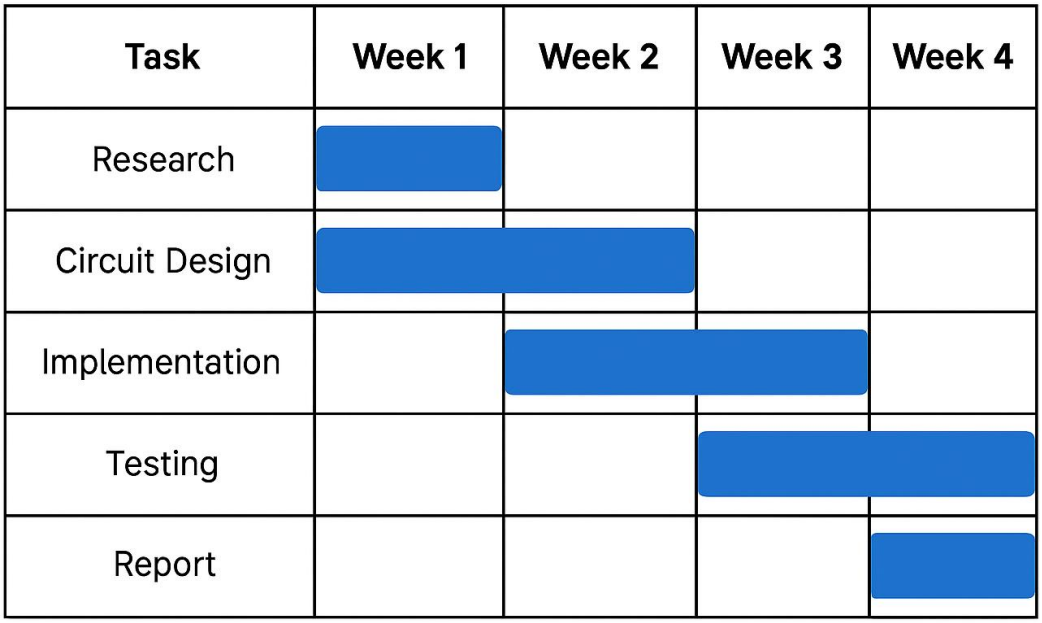
Gantt Chart

Automated Room Light Controller